ROI-Level Stage-vs-Rest Significance
mALFF

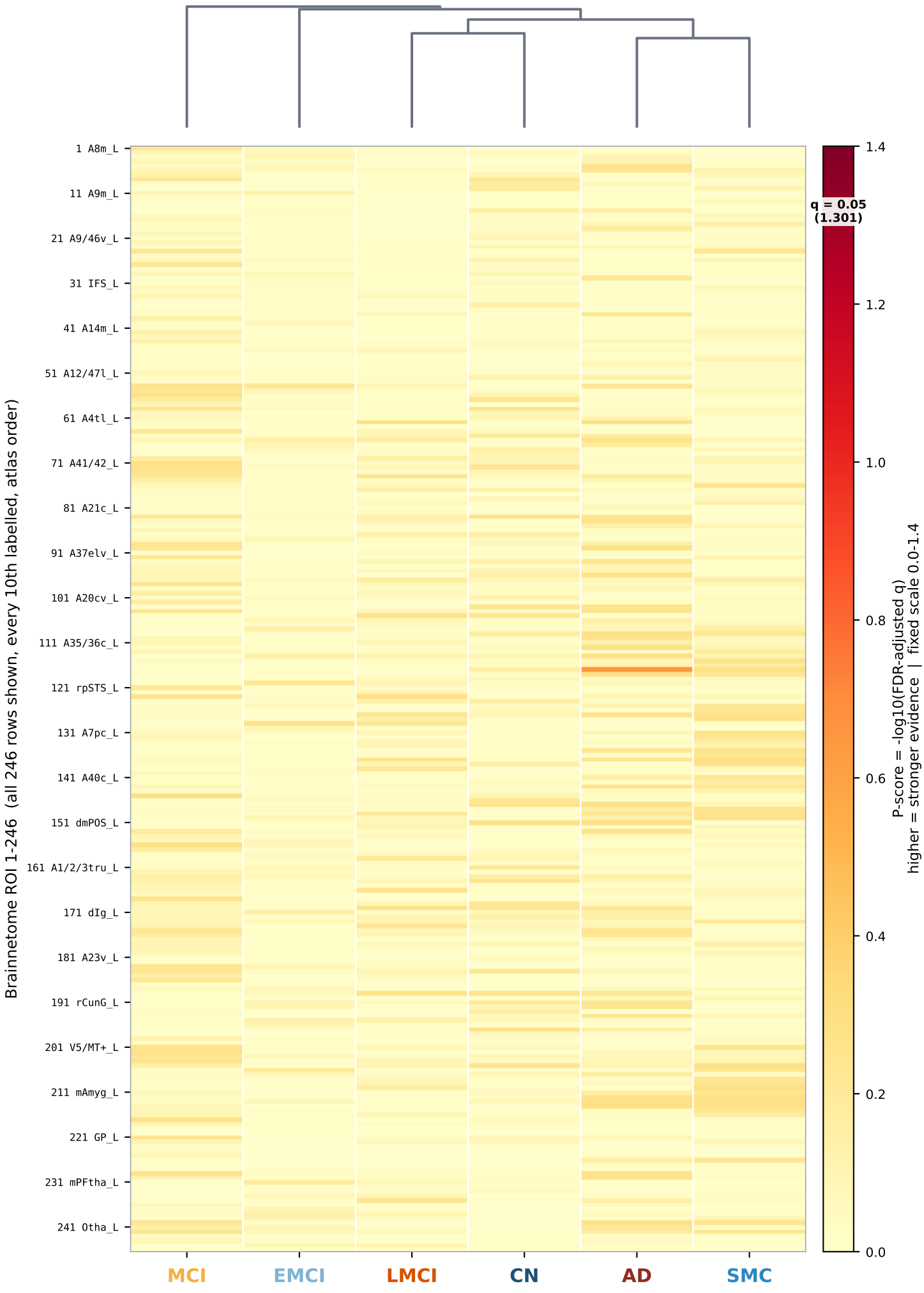
P = MULTIVARIATE group-level P-score, -log10(q) from the stage-vs-rest permutation test (24-test family)

max ROI P = largest value in that group's heatmap column -- DESCRIPTIVE, not a test

P = 0.738 **P = 0.012** **P = 0.119** **P = 0.154** **P = 0.738** **P = 0.403**

max ROI P 0.284 | max ROI P 0.259 | max ROI P 0.284 | max ROI P 0.284 | max ROI P 0.652 | max ROI P 0.284

clustering by P-SCORE PROFILE (correlation distance, average linkage, cophenetic r = 0.757)



columns ordered by the dendrogram above, NOT by disease stage

clinical progression order: CN -> SMC -> EMCI -> MCI -> LMCI -> AD

no ROI reaches q < 0.05: 0 of 1476 tests (highest P-score 0.652 < 1.301)

Heatmap: two-sided Mann-Whitney U per ROI per stage, Benjamini-Hochberg over 246 x 6 = 1476 tests per biomarker.
The tree measures similarity between STATISTICAL-SIGNIFICANCE PROFILES -- not raw biomarker activity, and not clinical progression.
No group-level P reaches 1.301 either; none of these annotations indicates a statistically significant result.